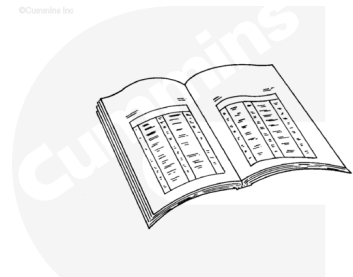


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

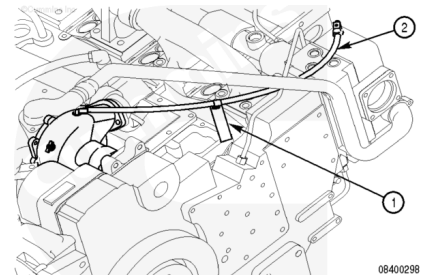
- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Open the draincock on the oil cooler housing and on the water pump.
- Remove the bypass tube clamp. Loosen both hose clamps. Remove the bypass tube, hose, and water connection.
- Disconnect the aftercooler supply hose from the water pump outlet. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)

- Remove the water pump outlet connection assembly.
- Remove the water transfer tube from the water connection.
- Remove and discard the two o-rings and the gasket.
- Remove the coolant filter. Refer to Procedure 008-006 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-006-tr.html>)
- Disconnect the lower radiator hose from the water pump inlet. Refer to Procedure 008-045 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-045.html>)

Remove

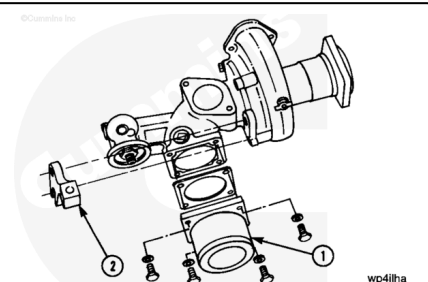
Rail Applications

- Remove the capscrews, nuts, washers, bracket, and p-clip (1).
- Remove the vent line (2) from the elbow fitting, p-clip, and t-union.



Remove the water inlet connection and gasket (1).

Remove the support bracket (2).

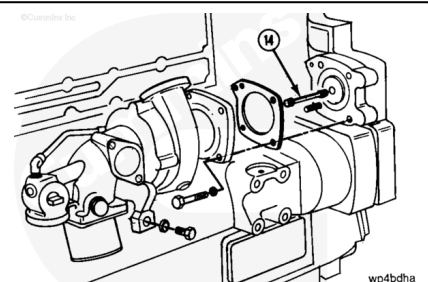


Remove the capscrew from the water pump support bracket.

Remove the three capscrews and the nut from the water pump mounting flange.

Remove the water pump and gasket. Discard the gasket.

Remove the drive shaft (14).

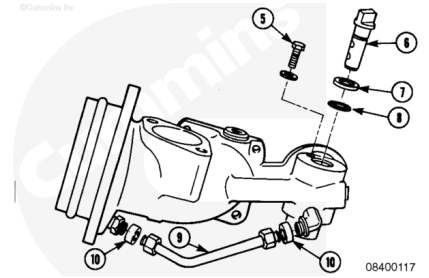


Disassemble

Remove the parts.

1. Shutoff valve capscrew
2. Water shutoff valve
3. Washer
4. O-ring
5. Water transfer tube (loosen the flare nut at each end)
6. Grommets.

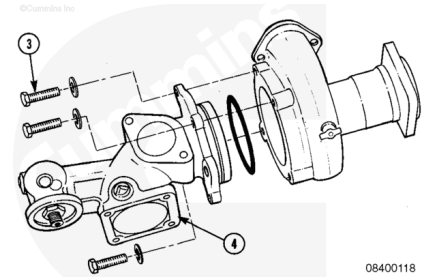
Discard the o-ring and grommets.



Remove the remaining capscrews (3).

Remove the inlet housing and o-ring (4).

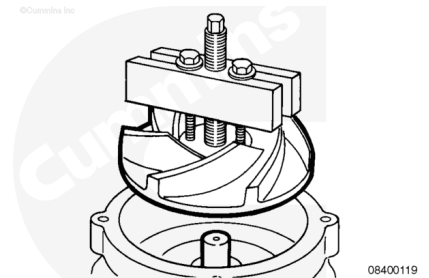
Discard the o-ring.



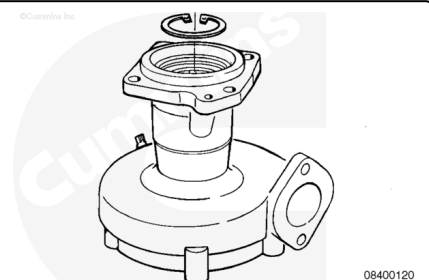
⚠ CAUTION ⚠

The jack screw in the puller must pass easily through the impeller bore to prevent damage to the impeller.

Use a standard puller, Part Number ST-647, or equivalent, and remove the impeller.



Remove the large retaining ring.

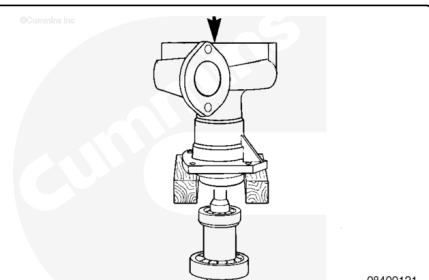


Support the water pump body, as shown.

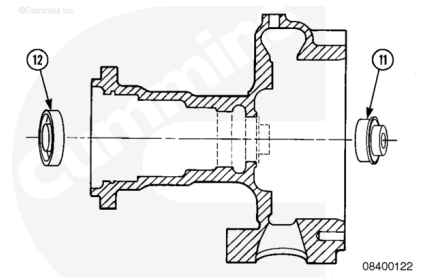
Use an arbor press to push on the impeller end of the shaft.

Remove the bearing and shaft assembly.

Discard the water pump seal seat.



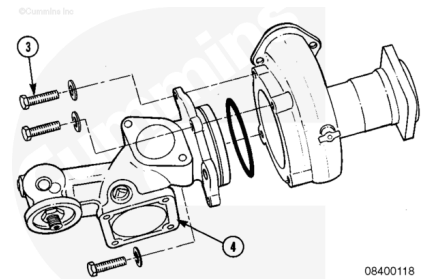
Remove and discard the water seal and oil seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



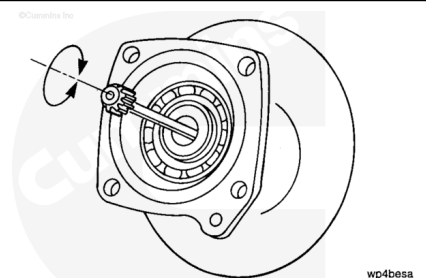
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean all the parts.

Dry with compressed air.

Rotate the shaft and inspect for rough or damaged bearings.



Inspect the drive shaft for wear. If the drive shaft splines are worn, check the female splines in the pump and in the water pump drive.

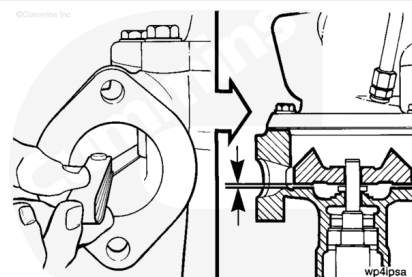


dw4shsa

Use a feeler gauge to measure the impeller-to-pump body clearance.

Impeller Vane-To-Body Clearance

mm		in
0.58	MIN	0.023
0.86	MAX	0.034

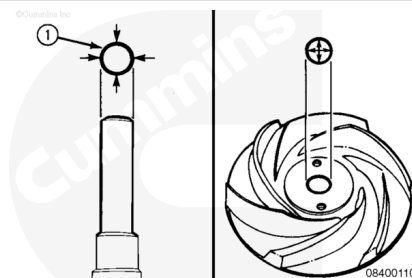


Use the following formula to determine the press fit.

Press fit = shaft outside diameter (1) minus impeller inside diameter.

Impeller to Shaft Press Fit

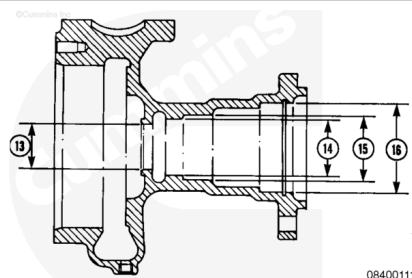
mm		in
0.03	MIN	0.001
0.07	MAX	0.003



Measure the inside diameter of the water pump housing bore.

Housing Bore Inside Diameter

	mm		in
(13) Water seal	36.45	MIN	1.435
	36.47	MAX	1.436
(14) Oil seal	44.43	MIN	1.749
	44.48	MAX	1.751
(15) Rear bearing	51.996	MIN	2.0471
	52.215	MAX	2.0557
(16) Front bearing	71.996	MIN	2.8345
	72.215	MAX	2.8431

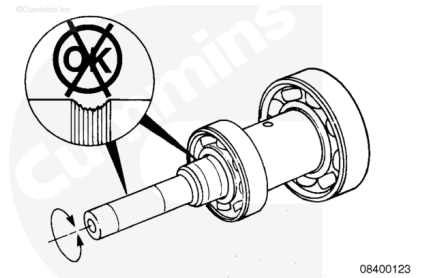


Inspect the bearing and shaft assembly for wear in the seal areas.

Spin the bearings to check for roughness.

If the bearings or shaft **must** be replaced, Refer to Procedure 008-071 in Section 8.

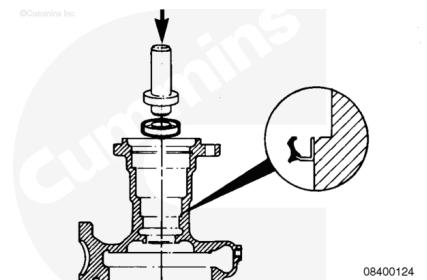
(/qs3/pubsys2/xml/en/procedures/20/20-008-071.html)



Assemble

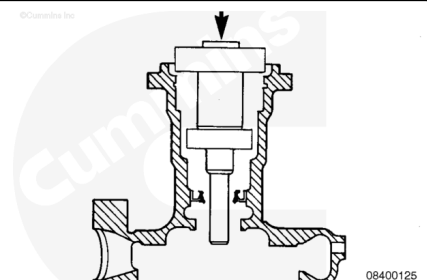
Use water pump seal driver, Part Number 3375320, to install the oil seal with the stepped lip facing down toward the impeller.

The seal **must not** be more than 0.51mm [0.020 in] below the top step in the water pump housing.



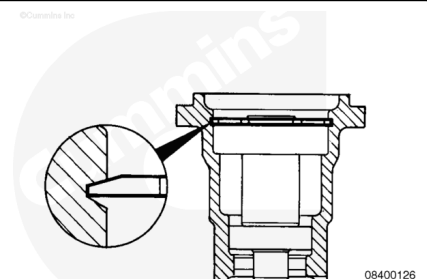
Support the water pump housing.

Use an arbor press and water pump bearing mandrel, Part Number 3375318, or equivalent, to push on the end of the shaft. Install the shaft and bearing assembly.



The beveled edge **must** be positioned as shown.

Install the large retaining ring.

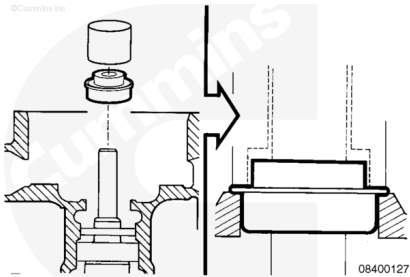


Use water pump seal driver, Part Number 3376091, or equivalent, to install the water seal.

The shoulder on the seal **must** touch the housing.

Note : This illustration shows the two-piece seal. This step is the same for the one-piece seal. The water

pump seal driver **must** be used to adjust the spring tension correctly.



⚠ CAUTION ⚠

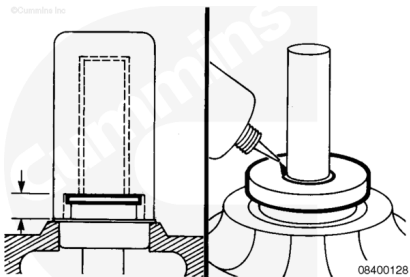
More than one drop of Loctite® 290, or equivalent, on the seal seat will glue the faces of the seal together resulting in damage to the seal.

Use an arbor press and water pump seal driver, Part Number 3376091, or equivalent, to install the seal seat.

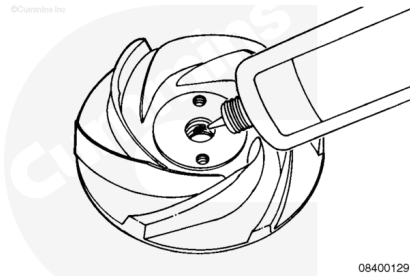
The seat dimension below applies to the one-piece and the two-piece seal. The seal tension will **not** be correct if the seat dimension is **not** within specifications.

Seat Dimension		
mm		in
10.52	MIN	0.414
10.57	MAX	0.416

Apply one drop of Loctite™ 290, or equivalent, to the seal seat, as shown.

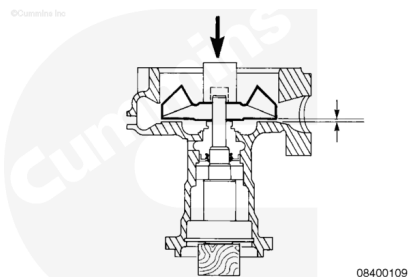


Apply a smooth coating of Loctite™ 609, or equivalent, to the inside diameter of the impeller, as shown.



Use an arbor press and a mandrel to install the impeller to the specified clearance.

Use a feeler gauge in the water outlet port to measure the clearance.



Impeller Vane-to-Body Clearance

mm		in
0.58	MIN	0.023
0.86	MAX	0.034

Install the washer (7).

Use vegetable oil to lubricate the o-ring (8).

Install the o-ring.

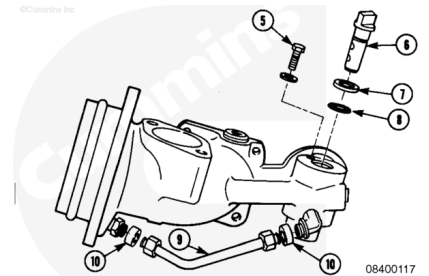
Install the shutoff valve.

Install the capscrew (5).

Torque Value: 20 n•m [177 in-lb]

Use vegetable oil to lubricate the new grommets (10).

Install the transfer tube.

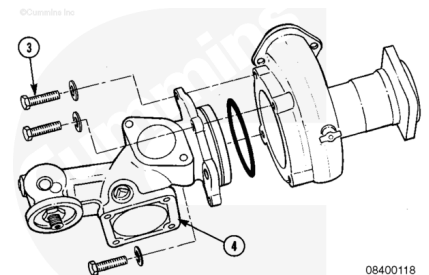


Use vegetable oil to lubricate the o-ring.

Install the parts, as shown.

Install the capscrews.

Torque Value: 45 n•m [33 ft-lb]

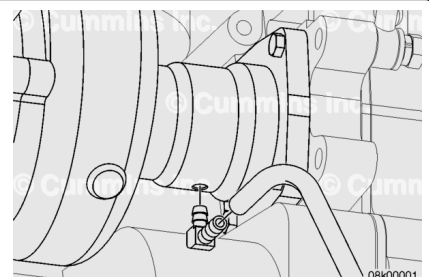


Coat the first two barbs of the 90 degree fitting with medium strength silicone sealant, Part Number 3823494, or equivalent.

Drive it into the weep hole of the pump until the shank touches the pump housing.

Do **NOT** put sealant on the tip of the barbed fitting, as this could cause contamination of the pump and/or plugging of the fitting upon installation.

Once the fitting is installed in the weep hole, install the hose by hand over the fitting until the end of the hose touches the shank of the fitting.



Install

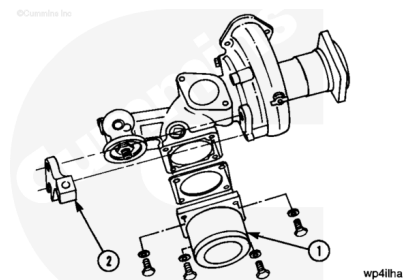
Install the inlet connection (1), gasket, and capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

Do **not** tighten the support bracket until the water pump is assembled to the engine. Use a heavy, plain washer on the capscrew that attaches through the slotted hole.

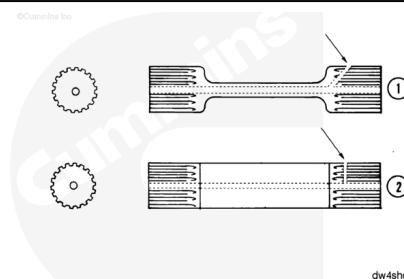
Install the support bracket (2).



Failure of the water pump drive shaft will result if the wrong shaft is installed.

Use shaft (1) with cast iron impeller pumps (marine applications). Use shaft (2) with phenolic (plastic) impeller pumps.

Install the shaft with the oil hole at the side of the shaft toward the water pump.



Use clean engine oil to lubricate the drive shaft (14).

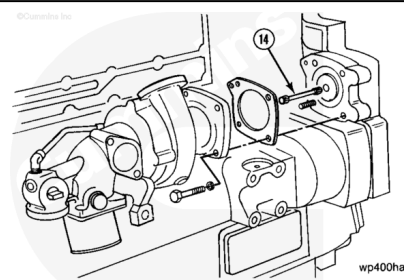
Install the shaft in the splined hole in the water pump drive.

Do **not** tighten the capscrews and nut until the support bracket is aligned with the cylinder block.

Install the gasket and water pump. Install three capscrews and nuts.

Tighten the capscrews and nuts.

Torque Value: 45 n•m [33 ft-lb]



The support bracket **must** be flat against the water pump and the cylinder block. Align the bracket before tightening the capscrews.

Install the capscrew through the bracket.

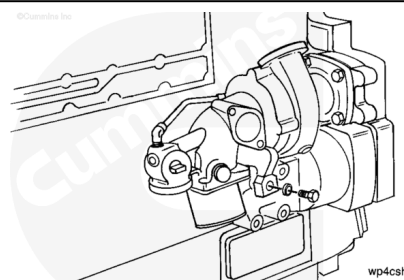
Tighten the capscrew.

Torque Value: 206 n•m [152 ft-lb]

Check the alignment between the support bracket and the water pump. Loosen the capscrew and adjust, if necessary.

Tighten the capscrews holding the support bracket to the water pump.

Torque Value: 45 n•m [33 ft-lb]



Rail Applications

Slide the p-clip over the loose end of the hose and align the bolt hole of the p-clip with the tapped boss at the lower right corner of the oil cooler housing.

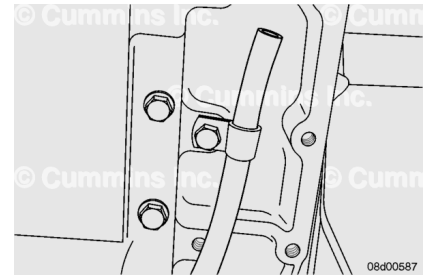
Secure the hose and p-clip to the block using the capscrew.

Tighten the capscrew.

Torque Value: 45 n·m [33 ft·lb]

Make sure **not** to twist or kink the hose.

If necessary, cut the hose to make sure the open end is unobstructed and in clear space.



Industrial Applications

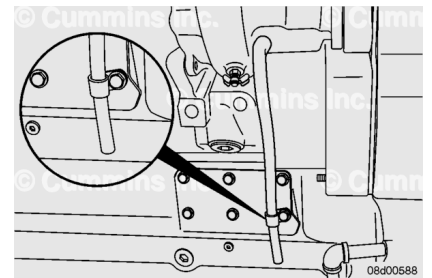
Secure the hose and p-clip to the hand hole cover using the hand hole cover capscrew.

Tighten the capscrew.

Torque Value: 45 n·m [33 ft·lb]

Make sure **not** to twist or kink the hose. The hose should go continuously down from the weep hole and its lowest point should be its open end.

If necessary, cut the hose to make sure the open end is unobstructed and in clear space.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

Use caution when filling coolant that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

- Connect the lower radiator hose. Refer to Procedure 008-045 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-045.html>)
- Connect the aftercooler supply hose to the water pump outlet connection. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Replace the coolant filter. Refer to Procedure 008-006 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-006-tr.html>)
- Close the draincock on the water pump and oil cooler housing.
- Connect the batteries or air starter supply line. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.